

DECK 02 · WEEK 1 · MORNING 2

Probability, Distributions & Portfolio Math

From moments of one return series to Monte Carlo pricing of multi-asset portfolios

CFA Level I · Vol 1 Learning Modules 3–6

Roadmap — Probability, Distributions, Portfolio Mathematics & Simulation

4 sections · ~2 hours · Built bottom-up from CFA readings

01 **Moments, shape & dispersion**
AM / median / mode · trimmed & winsorized · variance / MAD / semideviation · skew & kurtosis · correlation

02 **Probability trees & Bayes**
 $E[X]$, $\text{Var}[X]$ for discrete RVs · BankCorp EPS conditional tree · total probability rule · Bayes in credit & equity

03 **Portfolio risk & safety-first**
weighted return · covariance matrix · diversification · joint probability function · shortfall risk · Roy's criterion

04 **Lognormal & Monte Carlo**
lognormal asset prices · r_{cc} & $\sqrt{T}$ scaling · Astra volatility · 6-step Monte Carlo · bootstrap vs MC

Learning objectives

By the end of this deck you should be able to:

1. Compute and interpret the full set of descriptive statistics for an asset-return series — mean, variance / MAD / semideviation, skewness, kurtosis, covariance and correlation — and know which one answers which question.
2. Build a probability tree for a discrete random variable, read off conditional expected values and variances, and apply Bayes' formula to update a prior belief with new information.
3. Decompose portfolio variance into own-variance and covariance terms, enumerate how many distinct covariances an n-asset portfolio requires, and estimate covariance from a joint probability function when history is short.
4. Quantify shortfall risk using Roy's safety-first criterion and connect the SFRatio to the Sharpe ratio as a special case when the threshold equals the risk-free rate.
5. Explain why asset prices are modelled as lognormal while continuously compounded returns are modelled as normal, derive the $\sqrt{T}$ volatility-scaling rule, and walk through the 6-step Monte Carlo recipe for a derivative with no closed form.

SECTION 01

Moments, shape & dispersion

Mean, median and mode answer three different questions about the same return series — and outliers are what force you to pick carefully

Arithmetic mean is the most used — and the least robust to outliers. Trimmed and winsorized means reclaim robustness without abandoning the mean entirely.

$$\bar{X} = (1/n) \sum X_i$$

**median = middle-ranked value
most frequent value**

mode =

where: Median: position $(n+1)/2$ if n odd; mean of positions $n/2$ and $(n+2)/2$ if n even. Trimmed mean DROPS the tails; winsorized mean REPLACES them with the boundary percentile. Continuous return data usually has no single mode — quote the modal interval instead.

EAA Equity Index, 1,258 trading days	Observations	Mean	What it tells you
Arithmetic mean (all data)	1,258	0.035%	baseline — all outliers in
5% trimmed (cut below P2.5 / above P97.5)	1,194	0.048%	64 obs. dropped; mean rises
95% winsorized (caps at -1.934% / +1.671%)	1,258	0.038%	tails capped, not deleted

WHY IT MATTERS

Both robust means come out ABOVE the arithmetic mean — so the discarded observations were disproportionately large negatives. The gap (0.035% → 0.048%, +37% of the estimate itself) is a one-line diagnostic for influential left-tail outliers, before you plot anything.

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 89–93.

Variance and standard deviation describe two-sided dispersion — downside deviation and coefficient of variation sharpen the view when asymmetry matters

Target semideviation only penalises observations below a chosen threshold — the investor-relevant cut of risk.

$$s^2 = \Sigma(X_i - \bar{X})^2 / (n-1)$$

$$s_{\text{target}} = \sqrt{[\Sigma_{\{X_i \leq B\}}(X_i - B)^2 / (n-1)]}$$
$$CV = s / \bar{X}$$

where: B = target (minimum acceptable return). Note the denominator: n = TOTAL sample size, not the count of observations below B — the standard exam trap. CV is unitless, so it compares dispersion across datasets; it is meaningless when $\bar{X} \leq 0$.

CFA LM 3 EXAMPLES 3-4 — 12 MONTHLY PORTFOLIO RETURNS

Returns: 5, 3, -1, -4, 4, 2, 0, 4, 3, 0, 6, 5 (%). They sum to 27, so $\bar{X} = 27 / 12 = 2.25\%$.

1. Sample s : squared deviations from $\bar{X}$ sum to 96.25 $\rightarrow s = \sqrt{96.25 / 11} = 2.958\%$
2. Target $B = 3\%$: five months fall below; squared shortfalls sum to 84 $\rightarrow s_{\text{target}} = \sqrt{84 / 11} = 2.763\%$
3. Target $B = 2\%$: only four months fall below; sum drops to 53 $\rightarrow s_{\text{target}} = \sqrt{53 / 11} = 2.195\%$
4. $CV = 2.958 / 2.25 = 1.31 \rightarrow 1.3$ units of risk per unit of mean return

$$s = 2.958\% \quad \cdot \quad s_{\text{target}@3\%} = 2.763\% \quad \cdot \quad s_{\text{target}@2\%} = 2.195\% \quad \cdot \quad CV = 1.31$$

Skewness measures asymmetry and kurtosis measures tail weight — both matter because real asset returns are almost never normal

Normal = skew 0 and kurtosis 3 (excess kurtosis 0). Both formulas below are large-sample approximations — use them only when $n \geq 100$.

$$\text{Skew} \approx \frac{(1/n) \sum (X_i - \bar{X})^3 / s^3}{(1/n) \sum (X_i - \bar{X})^4 / s^4 - 3} \quad \text{Excess Kurtosis} \approx \frac{(1/n) \sum (X_i - \bar{X})^4 / s^4}{(1/n) \sum (X_i - \bar{X})^4 / s^4 - 3}$$

where: Cubing keeps the sign, so skew is signed; the 4th power kills the sign, so kurtosis only measures tail WEIGHT, never direction. Positive skew → long right tail (frequent small losses, rare large gains). Negative skew → the mirror.

NEGATIVE SKEW

Frequent small gains, rare large losses.

Typical of equity indices — crashes live in the left tail. Ordering: mean < median < mode.

LEPTOKURTIC • K_E > 0

More mass in BOTH tails and near the mean; less in the shoulders.

EAA Equity Index: skew -0.43, excess kurtosis +3.80 → fat-tailed and left-skewed at the same time.

IGNORE AT YOUR PERIL

A mean–variance model with skew -0.43 and K_E +3.80 in the data will systematically understate the odds of very bad and very good days. This is why parametric VaR gets backtested.

Covariance is scale-dependent and therefore hard to compare – correlation normalises it to the interval $[-1, +1]$ and becomes the workhorse

$r = +1$ is perfect positive LINEAR association. $r = 0$ means no LINEAR association — the two variables can still be perfectly related in a curve.

$$s_{XY} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1} \quad r_{XY} = s_{XY} / (s_X \cdot s_Y)$$

where: Same $n-1$ as the variance, and for the same reason — it makes s_{XY} an unbiased estimate of the population covariance. Covariance carries the units of $X \times Y$ (returns squared); correlation is unitless and bounded by ± 1 .

Pair · 24 monthly returns, decimals	Variance of each	Covariance	Correlation
Large-cap US vs world ex-US equity	0.001736 / 0.001488	0.001349	0.876
Intermediate vs long-term IG corporates	0.000174 / 0.000699	0.000318	0.951

READ THE TWO ROWS AGAINST EACH OTHER

The equity pair has $\sim 10\times$ the variances and $\sim 4\times$ the covariance of the bond pair — and the WEAKER relationship (0.876 vs 0.951). Covariance magnitude tells you about scale, not about strength of association. Only after dividing by $\sigma_X \sigma_Y$ can you compare pairs at all: that division is the entire job of the correlation coefficient.

Source: CFA Institute, Curriculum Vol 1, LM 3 pp. 119–121 (definitions) and LM 5 Example 2, pp. 165–166 (data).

Correlation captures only linear association — scatter plots remain essential to spot non-linearity, outliers, and spurious relationships

IT sector returns cluster tightly along a positively sloped line against the S&P 500; utility returns show no discernible pattern at all — same axes, opposite conclusions.

WHAT r CANNOT TELL YOU

1. Strength AND shape of the association
2. Curvature — a perfect parabola has $r \approx 0$
3. Data range, gaps, and clustering
4. Single outliers that manufacture (or destroy) the coefficient
5. Sub-populations / regime shifts hiding inside one cloud

ANSCOMBE'S QUARTET

Four datasets. Identical $\bar{X} = 9.00$, $s_X = 3.32$, $\bar{Y} = 7.50$, $s_Y = 2.03$, and identical $r = 0.82$.

I — roughly linear.

II — a clean curve.

III — perfectly linear except one outlier.

IV — X constant except one point.

Summary statistics do not describe data. Only the picture does.

¹ Spurious by chance: US retail sales of beer, wine and liquor vs atmospheric CO₂, 2000–2018, $r = 0.824$. No mechanism at all.

² Also spurious: two series both divided by market cap; or two series driven by a hidden third factor (height and vocabulary $\leftarrow$ age).

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 117–124, Example 7 (Anscombe 1973).

SECTION 02

Probability trees & Bayes

The expected value of a discrete random variable is the probability-weighted sum of its outcomes — variance is the probability-weighted squared deviation around it

Note the contrast with §1: there we averaged observations with equal weights (historical, sample). Here we weight OUTCOMES by probabilities — forward-looking, population. No $n-1$ anywhere.

$$E[X] = \sum P(X_i) \cdot X_i$$

$$\sigma^2[X] = \sum P(X_i) \cdot [X_i - E[X]]^2$$
$$\sigma[X] = \sqrt{\sigma^2[X]}$$

where: Probabilities must be mutually exclusive, exhaustive, and sum to 1. Order of operations is fixed: expected value first, then variance, then standard deviation.

CFA LM 4 EXAMPLE 1 — BANKCORP EPS FORECAST

Four possible EPS outcomes with probabilities 0.15, 0.45, 0.24, 0.16 and EPS \$2.60 / \$2.45 / \$2.20 / \$2.00.

1. $E[\text{EPS}] = 0.15(2.60) + 0.45(2.45) + 0.24(2.20) + 0.16(2.00) = \2.3405
2. $\sigma^2[\text{EPS}] = 0.15(2.60-2.34)^2 + 0.45(2.45-2.34)^2 + 0.24(2.20-2.34)^2 + 0.16(2.00-2.34)^2$
3. $= 0.01014 + 0.00545 + 0.00470 + 0.01850 = 0.03879$
4. $\sigma[\text{EPS}] = \sqrt{0.03879} \approx \$0.197 \rightarrow$ expected EPS of \$2.34 with a one-sigma band of roughly $\pm \$0.20$

$$E[\text{EPS}] = \$2.34 \quad \cdot \quad \sigma[\text{EPS}] = \$0.20$$

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 134–136.

A probability tree decomposes an unconditional forecast into scenario-conditional expectations that must recombine to the original number

Total probability rule for expected value: $E[X] = \sum E[X | S_i] \cdot P(S_i)$. The check that the recombined total matches the unconditional $E[X]$ is non-negotiable.

$$E[X] = E[X | S_1] \cdot P(S_1) + E[X | S_2] \cdot P(S_2) + \dots + E[X | S_n] \cdot P(S_n)$$

where: Scenarios $S_1..S_n$ must be mutually exclusive and exhaustive. Their probabilities sum to 1.

Scenario S_i	$P(S_i)$	$E[EPS S_i]$	Contribution	Branch probs
Declining interest rates	0.60	$0.25 \cdot 2.60 + 0.75 \cdot 2.45 = \2.4875	1.4925	0.15 / 0.45
Stable interest rates	0.40	$0.60 \cdot 2.20 + 0.40 \cdot 2.00 = \2.1200	0.8480	0.24 / 0.16
Unconditional $E[EPS]$	1.00	–	\$2.3405	= slide 11

THE TREE IS THE FORECAST – \$2.34 IS ONLY ITS SUMMARY

"If rates decline we see \$2.49; if they hold here, \$2.12" says far more than "\$2.34". And the branch probabilities must reconcile: $0.60 \times 0.25 = 0.15$, exactly the unconditional probability of a \$2.60 print on slide 11. If they do not tie, the forecast is internally arbitrage-able.

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 136–140.

Conditional variance is how scenario-conditioned forecasts keep track of residual risk inside each branch of the tree

Conditional variance is computed exactly like any variance — but with conditional probabilities and deviations from the CONDITIONAL mean, not the unconditional one.

$$\sigma^2[X \mid S] = \sum_i P(X_i \mid S) \cdot [X_i - E[X \mid S]]^2$$

where: Branch variances do not average up to the unconditional variance — the spread BETWEEN branch means is risk too (law of total variance, below).

DECLINING RATES · P = 0.60

$E[\text{EPS} \mid \text{declining}] = \2.4875

$$\sigma^2 = 0.25(2.60 - 2.4875)^2 + 0.75(2.45 - 2.4875)^2 = 0.00422$$

$$\sigma = \$0.065$$

STABLE RATES · P = 0.40

$E[\text{EPS} \mid \text{stable}] = \2.1200

$$\sigma^2 = 0.60(2.20 - 2.12)^2 + 0.40(2.00 - 2.12)^2 = 0.00960$$

$\sigma = \$0.098$ — the lower-mean branch is the riskier one too

THE TWO PIECES OF RISK MUST ADD BACK TO SLIDE 11

Within-branch risk: $0.60(0.00422) + 0.40(0.00960) = 0.00637$

Between-branch risk: $0.60(2.4875 - 2.3405)^2 + 0.40(2.12 - 2.3405)^2 = 0.03241$

Sum = $0.03879 = \sigma^2[\text{EPS}]$ on slide 11 — and 84% of it is WHICH scenario occurs, not what happens inside one.

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 138–140.

Bayes' formula reverses a conditional probability — it updates a prior belief by the ratio of likelihood to unconditional probability of the new information

The multiplier $[P(\text{Info} \mid \text{Event}) / P(\text{Info})]$ is the "evidence ratio". When it is >1 the posterior rises; <1 and the posterior falls.

$$P(\text{Event} \mid \text{Info}) = [P(\text{Info} \mid \text{Event}) / P(\text{Info})] \cdot P(\text{Event})$$

where: $P(\text{Event})$ = prior; $P(\text{Info} \mid \text{Event})$ = likelihood; $P(\text{Info})$ = unconditional probability; $P(\text{Event} \mid \text{Info})$ = posterior.

TECH-FIRM FRAMEWORK — READ THE UPDATE AS A FREQUENCY

A 500-stock index has 100 tech firms; 60 had returns $>10\%$. Of 400 non-tech firms, 100 had returns $>10\%$. What is $P(\text{tech} \mid R > 10\%)$?

1. Prior: $P(\text{tech}) = 100/500 = 0.20$ · unconditional $P(R > 10\%) = 160/500 = 0.32$
2. Likelihood: $P(R > 10\% \mid \text{tech}) = 60/100 = 0.60$
3. Bayes: $P(\text{tech} \mid R > 10\%) = (0.60 / 0.32) \cdot 0.20 = 1.875 \cdot 0.20 = 0.375$
4. Frequency check: of 160 firms with $R > 10\%$, 60 are tech $\rightarrow 60/160 = 0.375$ ✓

Posterior 0.375 = prior 0.20 × evidence ratio 1.875

The DriveMed expansion announcement lifts the implied probability of a consensus beat from 45% to 82% — a textbook Bayesian update

Three mutually exclusive events (beat / meet / miss) — the three posteriors must sum to 1, which is the final consistency check on the calculation.

Event	Prior	P(expansion Event)	Joint	Posterior
EPS exceeded consensus	0.45	0.75	0.3375	0.823
EPS met consensus	0.30	0.20	0.0600	0.146
EPS fell short	0.25	0.05	0.0125	0.030
P(expansion) · Σ posterior	1.00	—	0.4100	1.000

MECHANICS — TWO COLUMNS AND ONE DIVISION

Joint = prior × likelihood. P(expansion) = 0.41 is just the sum of the joint column — that is the total probability rule again. Posterior = joint / 0.41, e.g. 0.3375 / 0.41 = 0.823. One announcement moves the mass from 45 / 30 / 25 to 82 / 15 / 3, and the posteriors still sum to 1.000 — your check.

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, Example 4, pp. 144–147.

Bayes is most valuable where the base rate is low and the signal is imperfect — credit scoring, bankruptcy prediction, CEO turnover

Identical structure every time: prior $P(A)$ · likelihood $P(B|A)$ · unconditional $P(B)$. Note $P(B)$ is NOT a prior — it is the unconditional probability of the evidence, and it is the denominator.

CREDIT SCORE · TIMELY REPAYMENT

Prior:

$$P(A) = 0.90 \text{ pays on time}$$

Evidence:

$$P(B) = 0.80 \text{ good report}$$

Likelihood:

$$P(B | A) = 0.85$$

$$P(A | B) = (0.85 / 0.80) \times 0.90 \\ = 0.956$$

Ratio 1.06 → a weak signal. 90% → 95.6%.

BANKRUPTCY MODEL · FAILURE RATE

Prior:

$$P(\text{fail}) = 0.20$$

Evidence:

$$P(\text{good}) = 0.70$$

Likelihood:

$$P(\text{good} | \text{fail}) = 0.50$$

$$P(\text{fail} | \text{good}) = (0.50/0.70) \times 0.20 \\ = 0.143$$

Ratio 0.71 → posterior falls. 20% → 14.3%.

CEO FIRED · GOOD PERFORMANCE

Prior:

$$P(\text{fired}) = 0.05$$

Evidence:

$$P(\text{good}) = 0.50$$

Likelihood:

$$P(\text{good} | \text{fired}) = 0.30$$

$$P(\text{fired} | \text{good}) = (0.30/0.50) \times 0.05 \\ = 0.030$$

Low base rate dominates: 5% → 3%, still not 0.

SECTION 03

Portfolio risk

A portfolio's expected return is a simple weighted average, but its variance depends on pairwise covariances — which quickly dominate as n grows

For n assets the double sum has n^2 terms: n variances on the diagonal and $n^2 - n$ off-diagonal. At $n = 20$ that is 20 variances against 380 covariance entries — 19 co-movement terms for every own-risk term.

$$E[R_p] = \sum w_i E[R_i]$$

$$\sigma^2[R_p] = \sum_i \sum_j w_i w_j \text{Cov}(R_i, R_j)$$

where: Weights sum to 1. Diagonal $\text{Cov}(R_i, R_i) = \sigma^2[R_i]$. Off-diagonal terms capture co-movement.

CFA LM 5 — THREE-ASSET PORTFOLIO (S&P 500 / LT CORP BONDS / MSCI EAFE)

Weights 50% / 25% / 25%. $E[R] = 13\% / 6\% / 15\%$. Var diagonals 400 / 81 / 441 ($\%^2$). Off-diagonals 45 / 189 / 38.

1. $E[R_p] = 0.50(13) + 0.25(6) + 0.25(15) = 11.75\%$
2. Variance contribution (diagonals): $0.25 \cdot 400 + 0.0625 \cdot 81 + 0.0625 \cdot 441 = 132.625 (\%^2)$
3. Covariance contribution: $2 \cdot 0.50 \cdot 0.25 \cdot 45 + 2 \cdot 0.50 \cdot 0.25 \cdot 189 + 2 \cdot 0.25 \cdot 0.25 \cdot 38 = 63.25 (\%^2)$
4. $\sigma^2[R_p] = 132.625 + 63.25 = 195.875 \rightarrow \sigma[R_p] = 14.00\%$ (vs $\sqrt{132.625} = 11.52\%$ if all covariances were zero)

$E[R_p] = 11.75\%$ • $\sigma[R_p] = 14.00\%$ • co-movement costs ~250 bp of volatility

The covariance matrix captures everything portfolio math needs — for n assets it contains n variances plus $n(n-1)/2$ distinct covariances

The covariance matrix is symmetric ($\text{Cov}(A,B) = \text{Cov}(B,A)$), so only the upper (or lower) triangle needs to be estimated.

Inputs	Variances	Distinct covariances	Total
$n = 3$	3	3	6
$n = 5$	5	10	15
$n = 10$	10	45	55
$n = 20$	20	190	210
$n = 50$	50	1,225	1,275
$n = 100$	100	4,950	5,050

PRACTICAL IMPLICATIONS

Inputs grow as n^2 , so estimation becomes the bottleneck long before the formula does. A 500-stock portfolio needs $500 \times 499 / 2 = 124,750$ distinct covariances — with 10 years of monthly data you have only 120 observations per name.

Consequence: factor models (market plus style factors) and shrinkage estimators exist precisely to make the covariance matrix estimable and well-conditioned, not because anyone finds them elegant.

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, pp. 156–160.

A two-asset portfolio traces the efficient frontier — the curve bends left of the straight line, and that gap IS the diversification benefit

Stock 1: 4% return, 6% σ . Stock 2: 8% return, 15% σ . $\rho = 0.30$, so $\text{Cov} = 0.30 \times 6 \times 15 = 27$. Expected return moves in a straight line across the weights; σ does not.

$$\sigma^2[R_p] = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2 \cdot \rho \cdot \sigma_1\sigma_2 - \text{Cov} \quad w_1^* = (\sigma_2^2 - \text{Cov}) / (\sigma_1^2 + \sigma_2^2 - 2\text{Cov})$$

where: For any $\rho < 1$, $\sigma[R_p]$ sits strictly below the weighted average of σ_1 and σ_2 — the last column measures that gap. w_1^* is the minimum-variance weight.

w_1 / w_2	$E[R_p]$ (%)	$\sigma^2[R_p]$ (%) ²	$\sigma[R_p]$ (%)	vs. straight line
1.00 / 0.00	4.00	36.00	6.00	—
0.80 / 0.20	4.80	40.68	6.38	−1.42 pp
0.60 / 0.40	5.60	61.92	7.87	−1.73 pp
0.40 / 0.60	6.40	99.72	9.99	−1.41 pp
0.20 / 0.80	7.20	154.08	12.41	−0.79 pp
0.00 / 1.00	8.00	225.00	15.00	—
Min-σ 0.96/0.04	4.17	35.61	5.97	−0.42 pp

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, Example 1, pp. 159–160.

A joint probability function converts discrete scenarios into a forward-looking covariance — the right tool when history is short or regime-shifting

If X and Y are independent then $P(X,Y) = P(X) \cdot P(Y)$ and $\text{Cov}(X,Y) = 0$ — but the converse (zero correlation $\nRightarrow$ independence) does NOT hold.

$$\text{Cov}(R_A, R_B) = \sum_i \sum_j P(R_{A,i}, R_{B,j}) \cdot (R_{A,i} - E[R_A]) \cdot (R_{B,j} - E[R_B])$$

where: Sum over all discrete (i, j) outcomes weighted by their joint probability — the discrete analogue of the continuous covariance integral.

CFA LM 5 — BANKCORP × NEWBANK JOINT DISTRIBUTION

Three banking-industry states: good (0.20), average (0.50), poor (0.30). BankCorp returns 25 / 12 / 10%; NewBank returns 20 / 16 / 10%.

1. $E[R_A] = 0.20(25) + 0.50(12) + 0.30(10) = 14\%$ · $E[R_B] = 0.20(20) + 0.50(16) + 0.30(10) = 15\%$
2. Good: $(25-14)(20-15) = 55 \rightarrow$ weighted $0.20 \cdot 55 = 11$
3. Average: $(12-14)(16-15) = -2 \rightarrow$ weighted $0.50 \cdot (-2) = -1$
4. Poor: $(10-14)(10-15) = 20 \rightarrow$ weighted $0.30 \cdot 20 = 6$
5. $\text{Cov} = 11 - 1 + 6 = 16$. $\sigma_A = \sqrt{31} = 5.57$, $\sigma_B = \sqrt{13} = 3.61 \rightarrow \rho = 16 / (5.57 \times 3.61)$

$$\text{Cov}(R_A, R_B) = 16 (\% ^2) \quad \cdot \quad \rho(R_A, R_B) = 0.80 \quad - \quad \text{the number you can actually quote}$$

Shortfall risk — the probability of falling below a minimum acceptable return — is the downside metric that matches investor objectives

Symmetric σ treats upside and downside equally. Shortfall risk asks the only question clients care about: what is the chance I lose more than I can afford?

WHY SYMMETRIC RISK MISSES

Two portfolios with identical σ but different mean returns have very different shortfall probabilities under normality.

With target R_L , the portfolio with the higher mean has a smaller shaded area to the left of R_L — the same σ , but a dramatically different lived experience.

THREE PORTFOLIOS AT $R_L = 2\%$

Portfolio I: $E = 5\%$, $\sigma = 8\%$ $\rightarrow P(R < 2\%) = 37.7\%$

Portfolio II: $E = 8\%$, $\sigma = 8\%$ $\rightarrow P(R < 2\%) = 24.6\%$

Portfolio III: $E = 5\%$, $\sigma = 12\%$ $\rightarrow P(R < 2\%) = 41.7\%$

II beats I on the same σ with more return. II beats III on the same-ish return with less σ . Shortfall risk responds to BOTH — which is exactly why a single σ ranking is not enough.

Roy's safety-first criterion picks the portfolio that maximises the distance from the mean to the threshold, expressed in standard deviations

Under normal returns, $P(R_p < R_L) = N(-SFRatio)$ — so maximising the SFRatio minimises the shortfall probability. When $R_L = R_f$, SFRatio = Sharpe.

$$SFRatio = (E[R_p] - R_L) / \sigma[R_p] \quad \text{pick the portfolio with the highest SFRatio}$$

where: R_L = minimum acceptable return. For normal returns, shortfall probability = $N(-SFRatio)$.

CFA LM 5 EXAMPLE 3 — CAD 800,000 CLIENT · WITHDRAW CAD 30,000 IN 1Y

$R_L = 30,000 / 800,000 = 3.75\%$. Three allocations A / B / C with $E / \sigma = 25\% / 27\%$, $11\% / 8\%$, $14\% / 20\%$.

1. A: $SFRatio = (25 - 3.75) / 27 = 0.787$
2. B: $SFRatio = (11 - 3.75) / 8 = 0.906 \leftarrow \text{highest}$
3. C: $SFRatio = (14 - 3.75) / 20 = 0.513$
4. Shortfall probability for B under normality: $N(-0.90625) \approx 18.2\%$ (A: 21.6%, C: 30.4%)

Allocation B wins — the LOWEST expected return of the three, and still the safest against this threshold

SECTION 04

Lognormal & Monte Carlo

Asset prices are modelled as lognormal because they are bounded below by zero and right-skewed — properties the normal distribution cannot reproduce

Y is lognormal $\Leftrightarrow \ln Y$ is normal. Careful: the two parameters of a lognormal are the mean and variance of the OTHER distribution — of $\ln Y$, not of Y itself.

$$\begin{aligned} \text{median}[Y] &= \exp(\mu) & E[Y] &= \exp(\mu + 0.5\sigma^2) & \text{Var}[Y] &= \\ & \exp(2\mu + \sigma^2) \cdot [\exp(\sigma^2) - 1] \end{aligned}$$

where: μ and σ^2 are the mean and variance of the ASSOCIATED NORMAL, i.e. of $\ln Y$ — never of Y . This is the distribution Black–Scholes–Merton assumes for the underlying price.

BOUNDED BELOW BY ZERO

Stock prices cannot go negative. Normal distribution assigns probability to negative prices; lognormal does not.

RIGHT-SKEWED

Long right tail reflects compounding upside — a single year of +100% can double wealth, whereas -100% is the loss floor.

MEAN > MEDIAN

$E[\exp(X)] = \exp(E[X]) \times \exp(0.5\sigma^2) > \exp(E[X])$. Raise σ and the distribution spreads up but cannot spread below 0, so the mean is pushed right of the median. Jensen's inequality.

Continuously compounded returns are additive across time, so their sum over T periods scales linearly in mean and variance – the $\sqrt{T}$ rule for volatility

Independence + identical distribution (iid) is the assumption. When it breaks (autocorrelation, regime shifts), $\sqrt{T}$ scaling breaks too.

$$r_{0,T} = \sum r_t$$

$$E[r_{0,T}] = \mu \cdot T$$

$$\sigma^2[r_{0,T}] = \sigma^2 \cdot T$$

$$\sigma[r_{0,T}] = \sigma \cdot \sqrt{T}$$

where: Log returns ADD across time (simple returns compound), which is what makes this work. Mean scales with T, variance with T, so volatility scales with $\sqrt{T}$. Requires iid per-period returns with mean μ and variance σ^2 .

Horizon, given $\sigma_{\text{daily}} = 1.00\%$	T (days)	σ over T = $1\% \times \sqrt{T}$	Back-annualised: $\sigma_T \times \sqrt{(250/T)}$
1 trading day	1	1.00%	15.81%
1 week (5 trading days)	5	2.24%	15.81%
1 month (~21 trading days)	21	4.58%	15.81%
1 quarter (~63 trading days)	63	7.94%	15.81%
1 year (250 trading days)	250	15.81%	15.81%

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 181–183.

A week of Astra International closing prices lets us compute daily continuously compounded returns — $\sqrt{250}$ scaling gives an annualised volatility of 33.6%

5 prices give only 4 returns — nowhere near enough to estimate an expected return, but enough to run the volatility-annualisation pipeline end to end.

Day	Close (IDR)	$\ln(P_t/P_{t-1})$
Mon	6,950	—
Tue	7,000	+0.00717
Wed	6,850	-0.02166
Thu	6,600	-0.03718
Fri	6,350	-0.03862

ANNUALISATION PIPELINE

4 daily log returns $\rightarrow$ sample $\sigma \rightarrow \times \sqrt{250}$ to annualise.

1. Sum of returns = -0.090287, so $\bar{r} = -0.022572$ (ignore it — 4 obs.)
2. Sample variance $s^2 = 0.000452$, divisor $n - 1 = 3$
3. Daily $\sigma = \sqrt{0.000452} = 0.021261 = 2.13\%$
4. Annualised $\sigma = 0.021261 \times \sqrt{250} = 0.3362 = 33.6\%$

Annualised volatility $\approx 33.6\%$

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, Example 1, pp. 182–184.

Monte Carlo simulation values securities where no closed form exists by generating thousands of random price paths from assumed distributions

Specify once (steps 1–3), run repeatedly (steps 4–6). The more trials, the tighter the statistical estimate.

1 SPECIFY QUANTITY OF INTEREST

What are we valuing? E.g., C_{iT} = value of an Asian option at maturity given simulated stock-price path i .

2 SPECIFY TIME GRID

Divide the horizon into K sub-periods of length Δt . Example: 1 year with $K = 12$ gives monthly steps.

3 SPECIFY DATA-GENERATION METHOD

Distributional assumption for each risk factor (e.g., standard normal Z_k for stock shocks); pick μ , σ .

4 CONVERT RANDOM DRAWS TO PRICES

Apply the price-process model, e.g. $\Delta S = \mu S \Delta t + \sigma S Z_k$, iterated across the K sub-periods to build one path.

5 COMPUTE PAYOFF FOR ONE TRIAL

Calculate the claim's terminal payoff C_{iT} ; discount to C_{i0} using an appropriate rate. One complete trial is done.

6 REPEAT N TRIALS, SUMMARISE

Typically $N = 1,000$ to $100,000$ trials. Average of C_{i0} across trials is the Monte Carlo estimate of the claim's present value.

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 184–187.

Valuing an Asian option illustrates Monte Carlo end-to-end – average-minus-final payoff, 12 monthly steps, 1,000 trials

$Payoff = \max(S_T - \text{average}(S), 0)$. No closed form exists because the payoff depends on the path, not only the terminal stock price.

$$\Delta S = \mu \cdot S \cdot \Delta t + \sigma \cdot S \cdot Z_k \quad C_{iT} = \max(S_T - \bar{S}, 0)$$
$$\text{Value} \approx (1/N) \sum_i e^{(-rT)} \cdot C_{iT}$$

where: $Z_k \sim$ standard normal i.i.d. $\bar{S}$ = arithmetic average of the K simulated prices.

ILLUSTRATIVE RUN – 1,000 TRIALS, $T = 1Y$, $K = 12$

Draw 12,000 standard-normal shocks; build 1,000 paths of 12 monthly prices each; compute payoff, discount, average.

1. Path A: $S_T = \text{USD } 34$, $\bar{S} = \text{USD } 31 \rightarrow \text{payoff } \max(3, 0) = \text{USD } 3$
2. Path B: $S_T = \text{USD } 28$, $\bar{S} = \text{USD } 32 \rightarrow \text{payoff } \max(-4, 0) = \text{USD } 0$
3. Across the 1,000 trials: 654 finish at or below the average $\rightarrow$ zero payoff; 346 (34.6%) pay the difference
4. Discount every C_{iT} at an appropriate rate, then average the 1,000 present values $\rightarrow$ the option value today

Max simulated payoff USD 11 · value = mean of the 1,000 discounted payoffs, not of the payoffs

Bootstrap resampling swaps the assumed distribution for the historical one — same recipe, but draws with replacement from the observed sample

Bootstrap answers "what if history repeats?" Monte Carlo answers "what if my distributional assumption is correct?" Both are complements to closed-form pricing.

Dimension	Monte Carlo	Bootstrap
Source of random draws	Assumed distribution (e.g. normal)	Historical sample, with replacement
Distributional assumption	Required — must pick μ , σ , family	None — the sample IS the distribution
Strength	Any custom payoff or model structure	Real fat tails and skew, for free
Weakness	Wrong assumption → wrong answer	Cannot extrapolate beyond the sample
When to use	Derivatives pricing, stress testing	Confidence intervals, back-testing

THE ONLY STRUCTURAL DIFFERENCE IS STEP 3

Steps 1, 2, 4, 5 and 6 are identical to the Monte Carlo recipe. Step 3 alone changes: you draw WITH REPLACEMENT from the observed sample, each resample the same size as the original — so some observations appear twice and others not at all.

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 189–192.

Monte Carlo and bootstrap give only statistical estimates — analytical solutions remain preferable whenever they exist because they show cause-and-effect

More trials shrink sampling error only. They do nothing at all about model error — a wrong distribution simulated 10 million times returns a very precise wrong answer.

USE ANALYTICAL

- European options → Black–Scholes
- Vanilla bonds → DCF
- Normally-distributed portfolio VaR → parametric

Why:

- Exact
- Cheap
- Reveals cause-and-effect

USE MONTE CARLO

- Path-dependent payoffs (Asian, barrier, lookback)
- MBS with prepayment models
- Portfolio VaR with non-normal risk factors

Why:

- Flexible
- Handles any model
- Quantifies tail risk

USE BOOTSTRAP

- Confidence intervals on a complicated estimator
- Back-testing strategies over history
- Stress tests anchored in realised data

Why:

- No distribution assumed
- Fat tails preserved
- Simple to implement

But: cannot produce an event the sample never saw.

¹ Standard error of a simulated estimate falls as $1/\sqrt{n}$: halving it costs 4× the trials, and a 10× tighter band costs 100×.

² Analytical solutions show HOW parameters drive the answer (comparative statics). A simulation gives you a number and no derivative.

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 186–193.

Deck 2 — Probability, Distributions, Portfolio Mathematics & Simulation

■ Central tendency, dispersion, skewness, kurtosis and correlation are the five descriptive lenses every return series needs — each answers a different question.

■ Probability trees decompose an unconditional forecast into conditional ones and must recombine exactly. Total risk = risk within branches + risk between branches.

■ Returns are a weighted average; risk is not. $\sigma^2 = \sum_i \sum_j w_i w_j \text{Cov}$ — 20 assets means 20 variances against 190 distinct covariances, and estimation is the binding constraint.

■ Prices lognormal, log returns normal — mean exceeds median by $\exp(0.5\sigma^2)$. Volatility scales with $\sqrt{T}$ under iid; Astra: 2.13% daily $\rightarrow$ 33.6% annualised.

■ Sample statistics divide by $n-1$; probability-weighted expectations do not divide at all. Target semideviation keeps the $n-1$ but sums only shortfalls below B .

■ Bayes updates a prior by the evidence ratio $P(\text{Info} | \text{Event}) / P(\text{Info})$. DriveMed: 45% $\rightarrow$ 82%. With a low base rate, even a strong signal barely moves the posterior.

■ Roy's criterion maximises $\text{SFRatio} = (E[R_p] - R_L) / \sigma[R_p]$, which minimises $P(R < R_L)$ under normality. Set $R_L = R_f$ and it IS the Sharpe ratio.

■ Monte Carlo prices path-dependent claims by averaging discounted payoffs; bootstrap swaps the assumed distribution for the observed one. Both give estimates, never causes.